SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Seat No. | | | | | | |

Institute:	(0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Programme:	(070521) BACHELOR OF TECHNOLOGY(COMPUTER SCIENCE AND ENGINEERING)
Batch:	2023-27
Semester:	II
Course:	Statistics and Probability
Course Code:	0705210214

Date: 24/05/2024 — Maximum Marks: 45

Day: Friday — Time: 10:00 - 11:30

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. **Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

Que.1 Attempt any five parts of the following questions.

a)	Write the normal equations for linear regression.	3	CO1, BL2
b)	Write short note on cumulative distribution function corresponding to a random variable.	3	CO1, BL2
c)	Write short note on Bowley's coefficients of Skewness?	3	CO2, BL2
d)	Write short note on decile, quartile, percentile.	3	CO4, BL2
e)	Write short note on Quartile deviation.	3	CO4, BL1
f)	Write short note on Correlation.	3	CO6, BL1
g)	Define Poisson distribution and its properties.	3	CO6, BL1
h)	Write the condition under which a Binomial distribution can be approximated as Poisson Distribution.	3	CO6, BL1

Que.2 a) The mean of 5 observations is 4.4 and variance is 8.24. If three of the five observations are 1, 2 and 6, find the other two. — 5 CO2, BL4

b) Calculate first four moments about mean of the following distribution — 5 CO3, BL3

Observations:	0	1	2	3	4	5	6	7	8
Frequencies:	1	8	28	56	70	56	28	8	1

Que.3 a) Calculate the rank correlation coefficient for the following data — 5 CO4, BL4

X 68	64	75	50	64	80	75	40	55	64
Y 62	58	68	45	81	60	68	48	50	70

b) Fit both the straight-line curve (regression lines) with the help of the least-square method — 5 CO5, BL3

X	1	3	4	5	7	8
Y	12	16	8	10	14	16

Que.4 a) A random variable X has the density function given by — 5 CO6, BL3

$$f(x)=\begin{cases} kx^2, & 1\le x\le 2 \\ kx, & 2<x<3 \\ 0, & otherwise \end{cases}$$

. Find the constant k and probability $P(X<2.5)$

b) An urn contains 3 red and 7 white marbles. A marble is drawn from the bag and its color is observed. Then the marble is put back into the bag and the contents are thoroughly mixed. Find the probability that in 20 such drawings a red marble is selected at most 3 times. — 5 CO6, BL3

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Programme: (070521) BACHELOR OF TECHNOLOGY(COMPUTER SCIENCE AND ENGINEERING)

Batch: 2023-27
Semester: II
Course: Statistics and Probability
Course Code: 0705210214

Date: 03/12/2024 Maximum Marks: 45

Day: Tuesday Time: 13:30 - 15:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

Que.1 Attempt any five parts of the following questions.

a)	Write the expression for angle between two regression lines.	3	CO1, BL2
b)	Write short note on random variable and its probability distribution.	3	CO1, BL2
c)	Write short note on Karl Pearson's coefficients of Skewness?	3	CO2, BL2
d)	Write short note on Partition values.	3	CO4, BL2
e)	Write short note on mean deviation.	3	CO4, BL2
f)	Write short note on Regression.	3	CO6, BL2
g)	Define Binomial distribution and its properties.	3	CO6, BL1
h)	Write the condition under which a Binomial distribution can be approximated as Normal Distribution.	3	CO6, BL2

Que.2 a) The first group of two samples has 100 items with mean 15 and variance 9. If the whole group has 250 items with mean 15.6 and variance 13.44. Find the variance of the second group. 5 CO2, BL4

b) Using change of scale and change of origin calculate the mean and standard deviation for the following distribution 5 CO3, BL3

Ages (in Years):	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No of members:	3	61	132	153	140	51	2

Que.3 a) Calculate the rank correlation coefficient for the following data 5 CO4, BL4

X	75	50	64	80	75	40	68	64	55	64
Y	68	45	81	60	68	48	62	58	50	70

b) Calculate the correlation coefficient between X and Y and comment on their relationship 5 CO5, BL3

X	10	30	40	50	70	80
Y	2	6	8	10	14	16

Que.4 a) A random variable X has the following probability function: 5 CO6, BL3

X	0	1	2	3	4	5	6	7
$f(x)$	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

Find (i) k and (ii) $P(x < 6)$.

b) A factory produces inkpots, which are 10% defective. Find the probability that in a random sample of 400 inkpots produced by the machine 5 CO6, BL3

(i) Between 30 and 50
(ii) At the most 30
(iii) 55 or more

of the inkpots will be defective.

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Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY(COMPUTER SCIENCE AND ENGINEERING)

Batch: 2023-27,2024-28
Semester: I
Course: Statistics and Probability
Course Code: 0705210108

Date: 19/12/2024

Day: Thursday

Maximum Marks: 45

Time: 10:00 - 11:30

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

a. All questions are compulsory.
b. Write your seat number in words and figures in the Answer sheet provided to you.
c. Do not copy or help anyone in person or by using signs/ digital gadgets while you write the exam/ while you start the exam/ while you end the exam.
d. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
e. Make suitable assumptions wherever necessary.
f. Figures to the right indicate full marks.
g. A table of area under normal distribution will be provided during exam.
h. Non programmable calculator is allowed.

Que.1 Attempt any five parts of the following questions.

a)	Discuss the inclusive and exclusive form of data representation in tabular form.	3	CO1, BL1
b)	Write short note on decile, quartile, percentile.	3	CO1, BL1
c)	Write all the different coefficients of skewness.	3	CO2, BL1
d)	Define regression, and find the expression for angle between two regression lines.	3	CO4, BL1
e)	Explain the normal equations for the linear regressions for two variables.	3	CO4, BL2
f)	Explain Distribution function corresponding to continuous random variable. Write the expression for getting probability $P[1 < X < 5]$ in terms of distribution function.	3	CO6, BL2

g) Discuss the condition under which a Binomial distribution can be approximated as Poisson Distribution. 3 CO6, BL2

h) Define Poisson distribution and write its properties. 3 CO6, BL1

Que.2 a) Calculate Bowley's coefficients of Skewness for the following data 5 CO2, BL4

Observations:	0	1	2	3	4	5
Frequencies :	1	8	28	56	70	56

b) Obtain first four moments (about mean) of the following distribution hence conclude about the Kurtosis 5 CO3, BL3

Observations:	1	2	3	4	5	6
Frequencies :	4	5	20	40	50	65

Que.3 a) Calculate the correlation coefficient using short cut method only between the variables X and Y 5 CO4, BL4

X	10	12	14	18	20
Y	5	6	7	10	12

b) Fit both the straight-line curves (regression lines) with the help of the least-square method. 5 CO5, BL3

X	0	1	2	3	4
Y	1	1.8	3.3	4.5	6.3

Que.4 a) A bag contains 3 red and 5 white marbles. A marble is drawn from the bag and its color is observed. Then the marble is put back into the bag and the contents are thoroughly mixed. Using 5 CO6, BL3
(i) Binomial distribution
(ii) Normal approximation
Find the probability that in 20 such drawings a red marble is selected at most 3 times.

b) A factory produces components of which 1% are defective. The components are packed in boxes of 10. A box is selected at random. (i) Find the probability that the box contains exactly one defective component. (ii) Find the probability that there are at least 2 defective components in the box. 5 CO6, BL3

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Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY(COMPUTER SCIENCE AND ENGINEERING)

Batch: 2023-27
Semester: I
Course: Statistics and Probability
Course Code: 0705210108

Date: 29/05/2024 Maximum Marks: 45

Day: Wednesday Time: 10:00 - 11:30

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. **Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

Que.1 Attempt any **five** parts of the following questions.

a)	Write short note on Histogram, frequency polygon, Ogives.	3	CO1, BL3
b)	Write short note on primary and secondary data, describe with examples.	3	CO1, BL3
c)	Describe the different measures of dispersion.	3	CO2, BL2
d)	Describe the effect of change of origin and change of scale on regression coefficient?	3	CO4, BL2
e)	Define regression. Write down the properties of regression coefficient.	3	CO4, BL1
f)	Define continuous random variables and Explain p.d.f with its properties.	3	CO6, BL1
g)	Define Exponential distribution and its properties.	3	CO6, BL1
h)	Define (i) Mutually Exclusive events (ii) Exhaustive events.	3	CO6, BL1

Que.2 a) Using change of scale and change of origin calculate the coefficient of variation for the following distribution — 5 — CO2, BL4

Ages (in Years):	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No of members :	5	60	120	150	140	50	5

b) Obtain first four moments about origin of the following distribution hence find moments about mean — 5 — CO3, BL3

Observations:	1	3	4	5	7	8	10
Frequencies :	2	6	8	10	14	10	5

Que.3 a) Ten competitors in a musical test were ranked by the three judges A, B and C in the following order: — 5 — CO4, BL3

Ranks by A:	1	6	5	10	3	2	4	9	7	8
Ranks by B:	3	5	8	4	7	10	2	1	6	9
Ranks by C:	6	4	9	8	1	2	3	10	5	7

Using rank correlation method discuss which pair of judges has the nearest approach to common liking in music.

b) The table below is a data set on 8 test scores and hours spent studying for that test. Fit the straight-line curves (regression lines) with the help of the least-square method — 5 — CO5, BL3

Test Score:	95	90	89	82	80	70	60	95
Hours Spent:	6	4	5	5	4	3	0.5	6

Que.4 a) A biased dice is tossed 10 times and appearance of odd no or even no is observed with P (Even no) = 0.7 and P (Odd no) = 0.3. Let X be a random variable, which denotes the largest string of, even no appears. Find — 5 — CO6, BL3

(i) Probability distribution
(ii) Mean
(iii) Variance
(iv) Coefficient of skewness
(v) Coefficient of kurtosis

b) A bag contains two red and six white marbles. A marble is drawn from the bag and its color is observed. Then the marble is put back into the bag and the contents are thoroughly mixed. Using — 5 — CO6, BL3

(i) Binomial distribution
(ii) Normal approximation

Find the probability that in 20 such drawings a red marble is selected at most 3 times.

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Batch: 2023-27
Semester: I
Course: Statistics and Probability
Course Code: 0705210108

Date: 27/12/2023 Maximum Marks: 45

Day: Wednesday Time: 13:30 - 15:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. **Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**

Que.1		Attempt any **five** parts of the following questions.		
	a)	Write short note on Diagrams and its types?.	3	CO1, BL3
	b)	Write short note on grouped and ungrouped data, describe with examples.	3	CO1, BL3
	c)	Describe the different measures of central tendency.	3	CO2, BL2
	d)	Describe the effect of change of origin and change of scale on Karl person's correlation coefficient?	3	CO4, BL2
	e)	Define correlation. Write down the properties of correlation coefficient.	3	CO4, BL1
	f)	Define discrete random variables and Explain p.m.f with its properties.	3	CO6, BL1
	g)	Define Uniform Distribution and its properties.	3	CO6, BL1
	h)	Define (i) Equally likely events (ii) Independent event.	3	CO6, BL1

Que.2 a) Calculate Bowley's coefficients of Skewness (based on quartiles) for the following data — 5 — CO2, BL4

Observations:	0	1	2	3	4	5	6	7	8
Frequencies :	1	8	28	56	70	56	28	8	1

b) Obtain first four moments (about mean) of the following distribution hence conclude about the Kurtosis — 5 — CO3, BL3

Observations:	1	2	3	4	5	6	7	8	9
Frequencies :	4	5	20	40	50	65	30	10	5

Que.3 a) Calculate the correlation coefficient using short cut method only between the heights of fathers in inches (X) and their sons (Y) — 5 — CO4, BL4

X	65	66	57	67	68	69	70	72
Y	67	56	65	68	72	72	69	71

b) Fit the straight-line curves (regression lines) with the help of the least-square method. — 5 — CO5, BL3

X	65	66	57	67	68	69	70	72
Y	67	56	65	68	72	72	69	71

Que.4 a) A biased coin weighted so that P (H) = 3/7 and P (T) = 4/7, is tossed 20 times. Let X be a random variable, which denotes the largest string of, tells which occurs. Find — 5 — CO6, BL3

(i) Probability distribution
(ii) Mean
(iii) Variance
(iv) Coefficient of skewness
(v) Coefficient of kurtosis

b) A machine produces bolts, which are 10% defective. Find the probability that in a random sample of 400 bolts produced by the machine — 5 — CO6, BL3

(i) Between 30 and 50
(ii) At the most 30
(iii) 55 or more

of the bolts will be defective.

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